

OBJECT ORIENTED PROGRAMMING

#1

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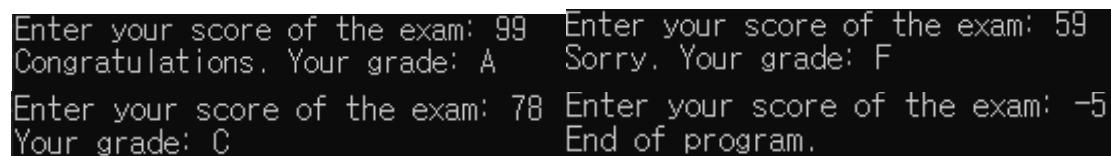
EXERCISE #1

```
#include <iostream>
using namespace std;

int main()
{
    int a;

    cout << "Enter your score of the exam: ";
    cin >> a;

    if (a > 0 && a < 60)
        cout << "Sorry. Your grade: F" << endl;
    else if (a > 59 && a < 70)
        cout << "Your grade: D" << endl;
    else if (a > 69 && a < 80)
        cout << "Your grade: C" << endl;
    else if (a > 79 && a < 90)
        cout << "Your grade: B" << endl;
    else if (a > 89 && a < 100)
        cout << "Congratulations. Your grade: A" << endl;
    else
        cout << "End of program." << endl;
    return 0;
}
```



The screenshot shows the program's output for three different inputs. The first input is 99, resulting in 'Congratulations. Your grade: A'. The second input is 59, resulting in 'Sorry. Your grade: F'. The third input is 78, resulting in 'Your grade: C'. The program also handles a negative input (-5) by displaying 'End of program.'

Input	Output
99	Enter your score of the exam: 99 Congratulations. Your grade: A
59	Enter your score of the exam: 59 Sorry. Your grade: F
78	Enter your score of the exam: 78 Your grade: C
-5	Enter your score of the exam: -5 End of program.

- This was a very easy basic code. I used if else condition to achieve the conditional of the grades,

EXERCISE #2

```
#include <iostream>
using namespace std;

char display(char c);
int display(int i);
double display(double d);

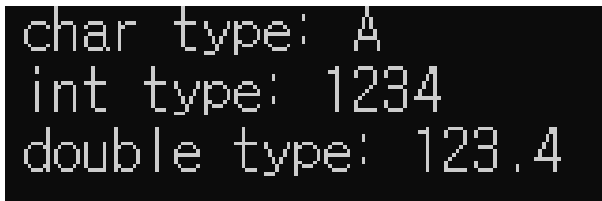
int main()
{
    display('A');
    display(1234);
    display(123.4);
}
```

```

        return 0;
    }

    char display(char c) { cout << " char type: " << c << endl; return 0; }
    int display(int i){ cout << " int type: " << i << endl; return 0;}
    double display(double d){ cout << " double type: " << d << endl; return 0;}

```



```

char type: A
int type: 1234
double type: 123.4

```

- This code was to overload a function. It has char, int, and a double type of function with the same name 'display'.

EXERCISE #3

```

#include <iostream>
using namespace std;

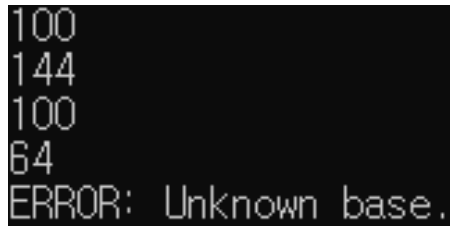
int cout_int(int a = 0, int b = 0);

int main()
{
    cout_int(100);
    cout_int(100,8);
    cout_int(100,10);
    cout_int(100,16);
    cout_int(100,20);
    return 0;
}

int cout_int(int y, int z)
{
    if (z == 0)
        cout << y << endl;
    else if (z == 8)
        cout << oct << y << endl;
    else if (z == 10)
        cout << dec << y << endl;
    else if (z == 16)
        cout << hex << y << endl;
    else
        cout << "ERROR: Unknown base." << endl;
    return 0;
}

```

```
}
```



```
100
144
100
64
ERROR: Unknown base.
```

- This code was to print a value in octal, decimal, and hexadecimal format using a function named 'cout_int'. I put a default value in the function parameter in case of error. So if a person forgot to put the format then it would print 'ERROR: Unknown base.'. This was achieved with the if else conditional statement.

EXERCISE #4

```
#include <iostream>
```

```
#include <cstdlib>
```

```
using namespace std;
```

```
int GetMaximum(int* x, int y);
```

```
int GetMinimum(int* x, int y);
```

```
double GetAverage(int* x, int y);
```

```
int main()
```

```
{
```

```
    int a;
```

```
    int* ptr = nullptr;
```

```
    cout << "# of data you want to input: ";
```

```
    cin >> a;
```

```
    ptr = new int[a];
```

```
    for (int i = 0; i < a; i++)
```

```
    {
```

```
        cout << i << ".Input an integer : ";
```

```
        cin >> *(ptr + i);
```

```
    }
```

```

        int max = GetMaximum(ptr, a);
        int min = GetMinimum(ptr, a);
        double avg = GetAverage(ptr, a);

        cout << "Minimum value: " << min << endl;
        cout << "Maximum value: " << max << endl;
        cout << "Average value: " << avg << endl;

        free(ptr);

        return 0;
}

int GetMaximum(int* x, int y){
    int max = x[0];

    for (int i = 1; i < y; i++)
    {
        if (x[i] > max)
        {
            max = x[i];
        }
    }

    return max;
}

int GetMinimum(int* x, int y){
    int min = x[0];

    for (int i = 1; i < y; i++)
    {
        if (x[i] < min)
        {
            min = x[i];
        }
    }

    return min;
}

```

```
double GetAverage(int* x, int y){
    double sum = 0.0;

    for (int i = 0; i < y; i++)
    {
        sum += x[i];
    }

    return sum / y;
}
```

```
# of data you want to input: 10
0.Input an integer : 100
1.Input an integer : -23
2.Input an integer : 1
3.Input an integer : 36
4.Input an integer : 123
5.Input an integer : 8
6.Input an integer : -12
7.Input an integer : 43
8.Input an integer : 98
9.Input an integer : 0
Minimum value: -23
Maximum value: 123
Average value: 37.4
```

- This was the most challenging code in this homework assignment. So, the program first asks for how much data they want to input, and the program allocates dynamic memory with the input number. Then the user has to enter the data. After that there are three different functions using a pointer and the variable to go through an array and find the maximum, minimum and the average value.

EXERCISE #5

```
#define _USE_MATH_DEFINES
#include <iostream>
#include <cmath>
using namespace std;

namespace shape {
    int rec_area(int a, int b) {
```

```

        return a * b;
    }

    int tri_area(int c, int d) {
        return (0.5 * c * d);
    }

    double cir_area(double e) {
        return M_PI * (e * e);
    }
}

int main()
{
    cout << "Area of Rectangle (W: 20, L: 10): ";
    cout << shape::rec_area(20, 10) << endl;

    cout << "Area of Triangle (b: 20, h: 10): ";
    cout << shape::tri_area(20, 10) << endl;

    cout << "Area of Circle (r: 20): ";
    cout << shape::cir_area(20) << endl;

    return 0;
}

```

```

Area of Rectangle (W: 20, L: 10): 200
Area of Triangle (b: 20, h: 10): 100
Area of Circle (r: 20): 1256.64

```

- We use 'shape' named namespace to use 3 functions to calculate the areas, `rec_area`, `tri_area`, and `cir_Area` which has the respective area calculating formulas inside. A quick way to operate is to just return the value that is calculated without defining it in the body of the function.